

A decorative graphic on the left side of the slide, consisting of a dark grey vertical band. Overlaid on this band is a light blue circuit board pattern. The pattern features several vertical lines of varying thickness, with numerous small circles (representing solder pads or vias) connected by thin horizontal and diagonal lines. The pattern is more dense on the left and tapers off towards the right.

SAM SERVELLON

TEAC 860

DRAGONBOX: A THREE PILLAR ANALYSIS

WHAT IS DRAGONBOX?



A mobile app designed to teach students how to solve algebraic linear equations through pattern recognition and logic puzzles



Utilizes gamification to avoid mentioning math concepts completely and presenting itself as a simple logic puzzle



Modifies the learning process into an organic problem-solving process using the properties of solving linear equations masked as game rules



Redefines the often-daunting task of solving linear equations as a game based on an image-based logic game



Allows students to activate problem-solving skills learned and honed from non-content related video games and tie a traditionally foreign concept to something familiar

An abstract graphic on the left side of the slide, consisting of a dark grey vertical band. Within this band, there is a complex pattern of thin, light blue lines that resemble a circuit board or a neural network. These lines are interconnected with small, light blue circles of varying sizes, creating a sense of connectivity and flow.

UNIVERSAL
DESIGN FOR
LEARNING

THREE PILLAR ANALYSIS



Design Multiple Means of Engagement



Access

Design Options for

Welcoming Interests & Identities

- Monolithic theme (Dragons & Underwater)
- Dragons & Fish are genderless
 - ↳ No overt identities within game
- Puzzle game with no real customization options

Support

Design Options for

Sustaining Effort & Persistence

- Each level is incremental and support is added until mastery is reached
 - ↳ Earn a star
- Can move onto the next chapter without earning all the stars but you get "boosts" for doing so

Executive Function

Design Options for

Emotional Capacity

- Individual game with no peer interaction
- Hints in the game are nonjudgmental and scaffolded



Design Multiple Means of Representation



Access

Design Options for Perception

- No real opportunities to customize the display of information
- Only one way to display/perceive information
- No real diversity of perspectives

Support

Design Options for Language & Symbols

- Good use of symbols & analogies to jump from picture tiles to math symbols
- Excellent decoding support
- Only support English & Spanish
- Does not address biases
- Uses both Visual and auditory media

Executive Function

Design Options for Building Knowledge

- Does connect old & new learning (requires teacher intervention to stick)
- Does an above average job of exploring patterns & relationships
- Doesn't really offer multiple pathways to understanding
- Game makers claim it maximizes transfer & generalization but does not do so as a stand-alone app



Design Multiple Means of Action & Expression



Access

Design Options for Interaction

- While the app does okay at interfacing with Apple's built in assistive tools, the game relies heavily on drag and drop and offers no closed captions
 - ↳ However, almost all audio is accompanied by pretty easily readable text

Support

Design Options for Expression & Communication

- Almost all audio is accompanied by text
 - ↳ However, the reverse is not true
- The structure of the game never changes, but new twists and puzzle types are introduced throughout
- Excellent graduated support but teacher intervention required to fully build fluency
- App assumes the user can read, hear, and has fine motor control

Executive Function

Design Options for Strategy Development

- Teacher would need to work with students to set goals outside of the app
- Not particularly adaptive
- Well organized & structured game
- No outside monitoring (this appears to have changed since they were acquired by Kahoot!)
- Does not challenge exclusionary practices? in fact, has some



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USABILITY
ASSESSMENT

THREE PILLAR ANALYSIS

CAN YOU USE IT? TEACHER EDITION



This particular app is no longer available since DragonBox was acquired by Kahoot!



New apps require a subscription model versus outright purchase but do offer some additional tools for classroom use such as progress tracking and alerts for the teacher



Only works on mobile devices so school would need to have iPads or students would have to log in on their personal devices



Only available in English and Spanish



The app now requires an internet connection

CAN YOU USE IT? (STUDENT EDITION)



App only works in landscape mode



Requires a medium degree of fine motor skills to drag and drop, tap, and flip puzzle tiles



Font used could be challenging for students who struggle with reading



App avoids colors that could be problematic for color-blind students



Students must be able to hear and see to use the app



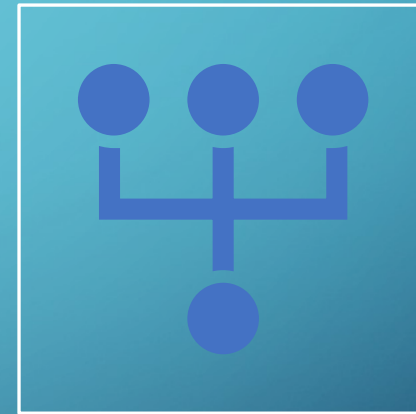
THEORETICAL
FOUNDATION
ANALYSIS

THREE PILLAR ANALYSIS

GAMIFICATION...IT'S ALL JUST A GAME

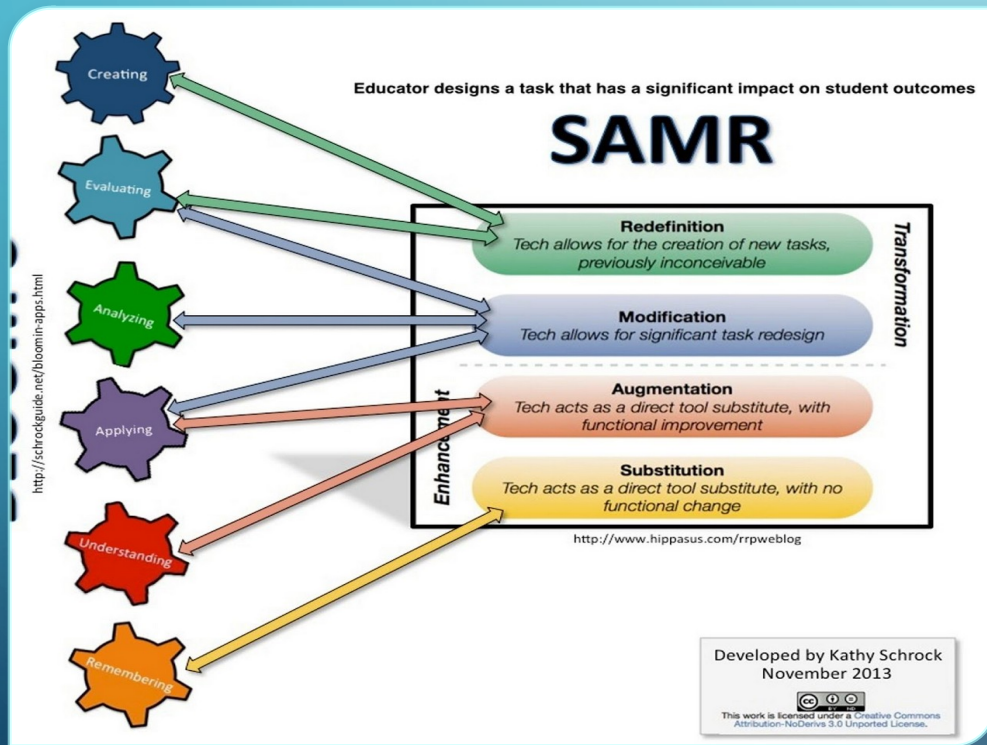


Gamification is the strategy of unlocking a student's growth mindset by incorporating gaming strategies and tactics to facilitate learning

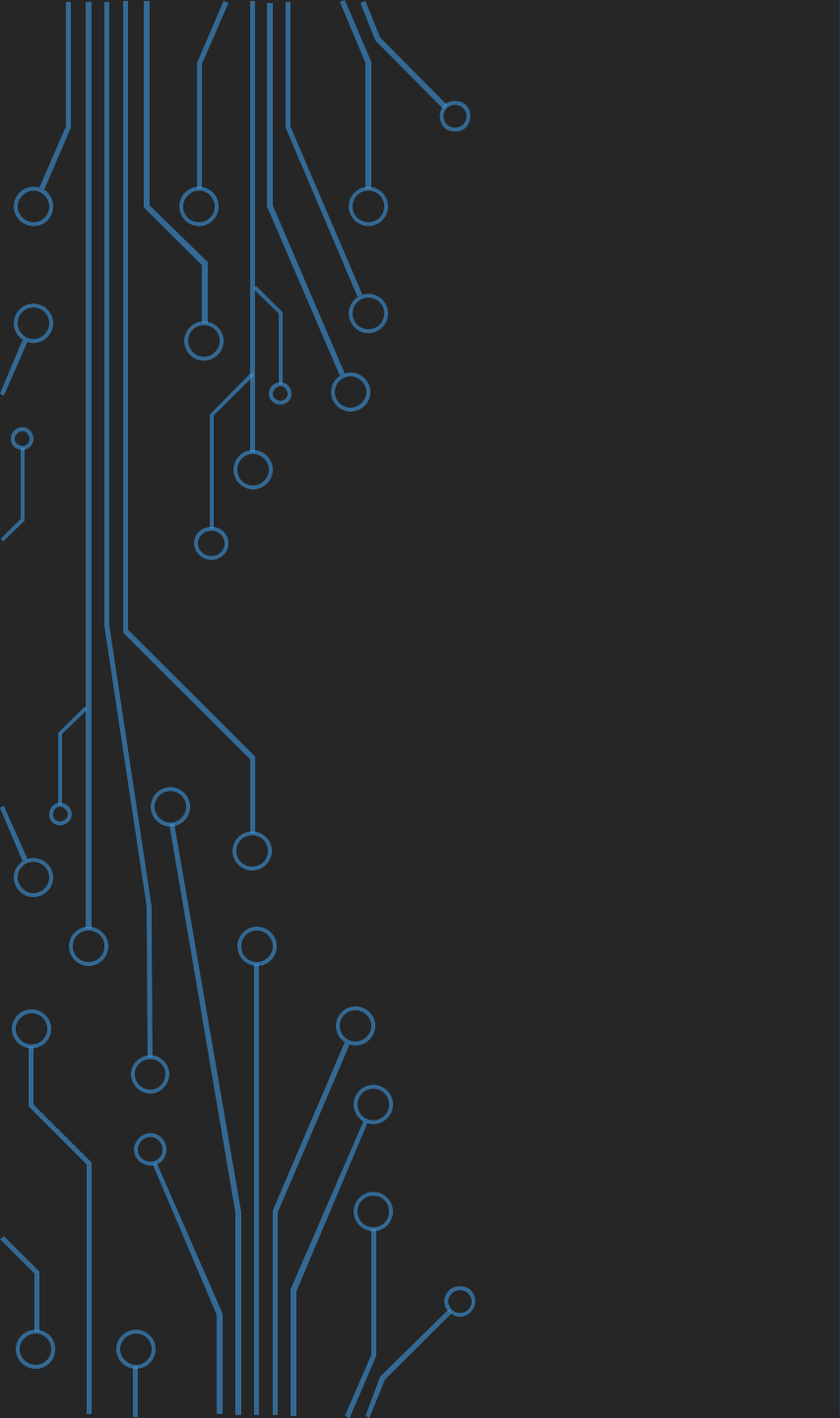


Dragonbox's creators state is a prime example of a successful implementation of gamification by leveraging Bloom's taxonomy and the SAMR model.

BUT IS IT GAMIFICATION?



- Gamification is the application of game mechanics in a non-gaming context
- DragonBox is a game, and arguably *just* a game without teacher intervention to help students transfer their skills in the game to actual learning

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PUTTING IT
ALL TOGETHER

IS THE APP WORTH IT?



Looking at the UDL assessment of the app through the app developers' claims of gamification, we see that while the app may have its uses, it is not the panacea it is presented to be (shocker)



App assumes a level of abled-ness and language proficiency that can be exclusive to many students



App also requires significant teacher intervention to transfer the game activities to real learning



Finally, the game is very linear and non-adaptive which goes against both many UDL principles but also the core of true gamification

IS THE APP WORTH IT? MAYBE

With all that said, I have used the game as a *supplement* to instruction and as a choice activity for students who have downtime in class with success



As with most (all, if we're being honest) educational technology tools, we need to critically assess the tools we are considering bring into our classrooms

What is our goal by bring this tool into our classroom?

What skills do my students have (or need) to use said tool?

What students are going to be (or feel) excluded with this too? Can we mitigate that?

Does this tool create extra work for me, the teacher, and is that extra work worth the potential gains?

RECOMMENDATIONS



Integrate accessibility tools available in both Android and iOS



Develop and curate teacher resources and lesson plans for use with the app



Develop a browser version of the game for schools that do not use iPads

FINAL THOUGHTS

DragonBox *can* be an incredibly useful tool as a support to learning in the classroom, but not a replacement for teaching

The app is well-made, well-packaged, and uses appealing buzzwords in its marketing but falls short of its primary claim of teaching students to solve linear equations in the self-contained app

You as the teacher need to decide if this type of tool fits into your classroom and teaching style and if it is appropriate for your students